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# Event Extraction from Course Announcements for Calendar Creation

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[https://github.com/Tim-UT/Event\\_Extraction](https://github.com/Tim-UT/Event_Extraction)

## Abstract

I built a system that turns Canvas course announcements into six calendar fields: classification, title, physical location, actionable URLs, start time, and end time. The final model combines a DistilBERT span tagger, TF-IDF classifiers, a temporal parser, and simple calendar rules. Mosaic v1.0 contains 5,823 fully labelled rows. On the final 443-row held-out test set, accuracy reaches 87.8% for classification, 88.3% for start time, and 92.3% for end time. All three target fields exceed the 85% project goal.

## 1 Introduction

Course announcements are messy: one post may contain several dates, relative time phrases, links, event titles, and words such as Zoom or Canvas that are not physical locations. The goal is to turn each announcement into a useful calendar record without losing the original evidence. Machine learning helps interpret phrases in context, while a small set of rules keeps dates and times in a valid calendar format.

## 2 Illustration and Related Work

Figure 1 shows the final pipeline. Transformers introduced self-attention (Vaswani et al., 2017); BERT and DistilBERT provide contextual encoders at different computational costs (Devlin et al., 2019; Sanh et al., 2019). BeeSL treats event extraction as sequence labelling (Ramponi et al., 2020), while Text2Event and UIE generate structured outputs from text (Lu et al., 2021, 2022). OmniEvent provides a modular event-understanding toolkit (Gao et al., 2023). This project uses a smaller, inspectable schema because calendar records need strict output rules.

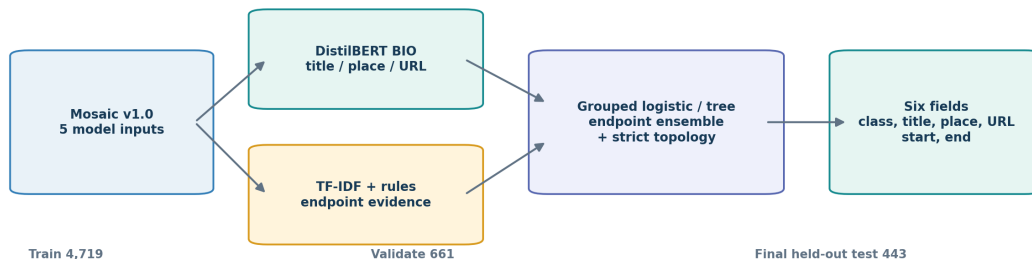


Figure 1: Final hybrid pipeline and held-out evaluation path.

### 3 Data Processing

The data came from Canvas exports. I preserved paragraph boundaries and complete link destinations, split multi-item announcements into item-level rows, reviewed the labels, and replaced identifying information with stable aliases. The same aliases are used inside the body text. Only rows with a complete, defensible set of labels were kept for modelling. The time rules are fixed: not related has no times, a notice starts at its post time, a deadline has an end time, and an event has both start and end times. For example, “Tuesday 5–6 P.M.” is labelled as an event with start 17:00 and end 18:00.

I split the data by source group so that related rows cannot appear in both development and test data. There is no source-group or exact-input overlap across splits. Courses may appear in more than one split because the intended use is to process future announcements from known courses. Table 1 summarizes the data used for model development and final testing.

Table 1: Mosaic v1.0 model-development and final-test partitions.

| Split      | Rows  | Courses | Groups | Not related | Notice | Deadline | Event |
|------------|-------|---------|--------|-------------|--------|----------|-------|
| Train      | 4,719 | 62      | 2,533  | 384         | 1,573  | 938      | 1,824 |
| Validation | 661   | 52      | 446    | 89          | 318    | 130      | 124   |
| Final test | 443   | 47      | 327    | 27          | 262    | 93       | 61    |

### 4 Architecture and Training

The five input fields are joined with field markers. DistilBERT reads up to 256 tokens and tags title, location, URL, date, and time spans. Word- and character-level TF-IDF models handle classification and time-endpoint evidence. Logistic and ExtraTrees models combine grouped out-of-fold predictions, while validation selects the regularization, ensemble weights, and endpoint thresholds. A deterministic parser normalizes dates and times, then the class-time rules enforce the final calendar structure. DistilBERT was trained for four epochs with AdamW (learning rate  $2 \times 10^{-5}$ ), batch size 8, and gradient accumulation over two steps.

Training uses only the training and validation files. Validation selects the model settings and thresholds; the final-test labels are not used during development.

### 5 Baseline Model

The baseline applies fixed rules in order. A date with deadline words predicts a deadline; a dated calendar term with a time range predicts an event; other dated course actions become notices; everything else is not related. Regular expressions extract full URLs, physical room patterns, dates, and times, while the announcement heading supplies a title candidate. It uses the same output formatting as the final model but has no learned context.

### 6 Quantitative Results

Table 2 and Figure 2 compare the baseline with the final hybrid. The hybrid exceeds the 85% target for classification and both temporal endpoints. Classification improves by 54.4 percentage points, start-time accuracy by 59.1 points, and end-time accuracy by 9.9 points. Actionable-URL accuracy is also strong at 97.1%, with complete query strings preserved. Classification macro-F1 is 81.8%. On rows with a labelled time, exact match is 87.0% for start time ( $n = 323$ ) and 89.6% for end time ( $n = 154$ ), showing that the temporal scores do not come only from blank fields.

Table 2: Strict exact-match accuracy on the final held-out test set ( $n = 443$ ).

| Field             | Rule baseline | Final hybrid |
|-------------------|---------------|--------------|
| Classification    | 33.4%         | 87.8%        |
| Physical location | 88.0%         | 73.1%        |
| Actionable URLs   | 96.8%         | 97.1%        |
| Extracted title   | 6.8%          | 8.6%         |
| Start time        | 29.1%         | 88.3%        |
| End time          | 82.4%         | 92.3%        |

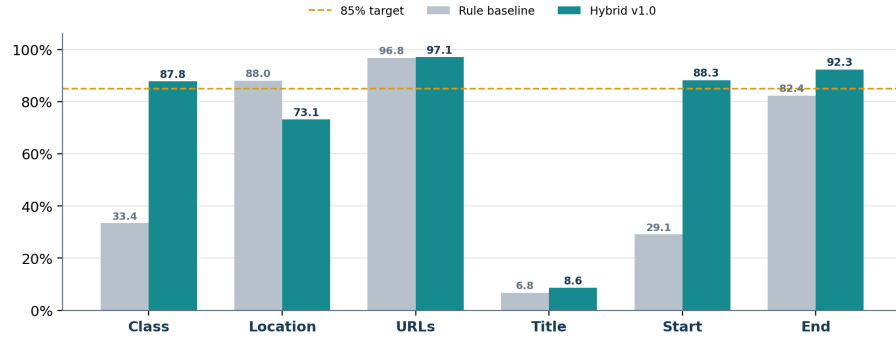


Figure 2: Baseline and final hybrid exact-match accuracy.

## 7 Evaluation on Data Not Used for Training

The final test set contains 443 announcements from 327 source groups. It was selected using annotation quality only, without using model predictions or correctness, and complete source groups were kept together. After the split was fixed, the test labels were not used for training, model selection, or threshold setting, and the test was scored once. There is no source-group or exact-input overlap with development data. These are unseen groups from the same collection rather than announcements gathered later. The model still meets the project goal on this held-out set.

## 8 Qualitative Results

**Exact temporal success.** For “The plenary lecture (Tuesday 5–6 P.M.),” both the label and prediction are an event from 17:00 to 18:00. The system correctly connects the weekday to the date and normalizes the range.

**Semantic class error.** A weekly orientation post is labelled as a notice but predicted as not related. The difficult part is deciding whether general course information belongs on a calendar.

**Class and title errors.** For “Extra Midterm Office Hours,” the correct label is an event, but the model predicts a notice and uses the post time. In another example, the event time is correct but the title is too long. These cases show the main class and span errors.

## 9 Discussion

The results show that data preparation mattered as much as model choice: each row needs the text that supports its label. The class-time rules prevent impossible records and set notices to their post time. Macro-F1 is lower than accuracy because the small not-related class is harder than the common notice class. The baseline’s 88.0% location score is misleading: 395 of 443 location labels are blank, and it matches none of the 48 nonblank locations. The hybrid finds some locations, but location and title extraction remain weak because a short span-boundary error makes the entire exact-match prediction wrong. More examples near the not-related/notice boundary and more labelled physical locations should give the largest improvement.

## **10 Ethical Considerations**

Canvas announcements are educational records, so raw exports and the alias mapping must remain private. Mosaicing removes direct identifiers, but uncommon details may still reveal identity. Because a wrong date can cause a missed event, the system should show its source text and ask for confirmation before calendar insertion.

## **11 Difficulty, Quality, and Conclusion**

The main difficulty is that one announcement can contain several actions and dates, relative times depend on the post timestamp, and all six outputs must describe the same item. This requires more than a single text classifier: the final system combines item-level reviewed data, contextual span extraction, temporal parsing, and calendar constraints. It exceeds 85% on classification, start time, and end time on held-out data, showing that course announcements can be converted into useful calendar records.

I used ChatGPT for language editing and code troubleshooting; I reviewed the data, experiments, and reported results myself (OpenAI, 2026).

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